

The Fidelity–Utility Paradox in Surrogate-Based Reinforcement Learning for HVAC Control

1 · Predictive fidelity

Two fast surrogates of live BOPTEST:

BB black-box — coarse, smooth
 $\text{RMSE}_T = 1.557^\circ\text{C}$

GB physical twin — *more accurate*
 $\text{RMSE}_T = 0.644^\circ\text{C}$

2 · Fidelity–utility paradox

Train PPO on each; score live BOPTEST:

accurate **GB** → **collapse** $m_s > 1$

coarse **BB** → **usable** $m_s \approx 0.08$

Higher prediction accuracy does not imply better closed-loop control.

3 · Role-separated hybrid

BB rollout + frozen **GB** reward censor

→ **robust** live control:

$m_s = 0.041$

<5% comfort violation

~85× live-loop throughput

Takeaway: evaluate HVAC surrogates by downstream *control utility*, not by prediction accuracy alone.

Transfer boundary: the calibration pipeline transfers to new hydronic cases; a frozen policy does not uniformly transfer.